

Statistics Primer for Whitespace 2

Originality Program — Whitespace 2 Empirical Decomposition

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1 Introduction

This primer accompanies the Phase 0.1 scoping plan for Whitespace 2, an empirical time-series study testing whether demographic plurality and semantic plurality of scientific output have diverged over 1970–2024. The plan commits to specific statistical methods, some of which are standard in scientometrics but may be unfamiliar outside that literature. This document explains each committed method from first principles: what question it answers, its formal definition, the intuition behind it, when it works and when it fails, and how Whitespace 2 uses it.

The organization follows the structure of the analysis pipeline:

- **Part I** — diversity metrics used for the demographic, semantic, and canonical-concentration time series (Sections 2–6).
- **Part II** — the four co-primary divergence tests committed in Phase 0.2 pre-registration (Sections 7–10).
- **Part III** — break-point analysis for locating when any divergence begins (Sections 11–12).
- **Part IV** — auxiliary methods used across multiple tests: bootstrap confidence intervals, multiple-comparisons correction, diachronic embedding alignment, and multicollinearity diagnostics (Sections 13–16).

Each section is self-contained. Read linearly for a full build-up, or jump to specific methods as needed. A glossary of symbols is in the appendix.

Part I

Diversity Metrics

2 Shannon Entropy

Question answered. Given a categorical distribution (e.g., authors’ countries, paper cluster assignments), how “spread out” is the mass across categories?

Formal definition. For a distribution with probabilities $p_1, \dots, p_K$ over K categories:

$$H(p) = - \sum_{i=1}^K p_i \log p_i$$

By convention $0 \log 0 = 0$. The base of the log determines the units: base-2 yields bits, base- e yields nats, base-10 yields hartleys. Whitespace 2 uses natural log (nats); the choice affects units but not conclusions.

Intuition. Shannon entropy is the expected “surprise” of a random draw. If one category dominates ($p_1 \rightarrow 1$), drawing that category is unsurprising and $H \rightarrow 0$. If all categories are equally likely ($p_i = 1/K$), each draw is maximally surprising and $H = \log K$.

The quantity $\exp(H)$ is the *effective number of categories* — the number of equally-common categories that would produce the same entropy. If papers are distributed over 50 cluster IDs but half the mass is in 3 clusters, $\exp(H)$ might be around 8: roughly equivalent to being spread over 8 equally-common clusters.

Finite-sample bias. The naïve “plug-in” estimator $\hat{H} = - \sum \hat{p}_i \log \hat{p}_i$ systematically underestimates the true entropy. The leading-order bias is

$$\mathbb{E}[\hat{H}] - H(p) \approx -\frac{K-1}{2N}$$

where N is the sample size. For $K = 100$ and $N = 1000$, the bias is about 0.05 nats—non-negligible. The *Miller-Madow bias correction* adds $(K-1)/(2N)$ to the plug-in estimate:

$$\hat{H}_{\text{MM}} = \hat{H} + \frac{K-1}{2N}$$

Whitespace 2 uses this correction on all Shannon entropy estimates.

When it fails.

- With many rare categories ($K \sim N$), even Miller-Madow underestimates. Consider Chao-Shen or NSB estimators in that regime (not used in ws2 primary analyses).
- Requires a discrete distribution. For continuous features (career stage in years), discretize into bins first; bin choice affects the estimate.

Use in ws2. Per-dimension demographic diversity (gender, country, institution type, prestige tier, career stage bin, training institution) uses Shannon entropy as the primary metric. Cluster entropy (Section 6) is Shannon entropy applied to paper-embedding cluster assignments.

3 Gini-Simpson Index

Question answered. Given a categorical distribution, what is the probability that two independently drawn samples fall into different categories?

Formal definition. For probabilities $p_1, \dots, p_K$:

$$\text{GS}(p) = 1 - \sum_{i=1}^K p_i^2$$

The Simpson index $D = \sum p_i^2$ is the probability that two draws match; Gini-Simpson is its complement.

Intuition. If a single category dominates, two random draws likely match, $D \rightarrow 1$, and $\text{GS} \rightarrow 0$. If categories are equal, the match probability is $1/K$ and $\text{GS} = 1 - 1/K \rightarrow 1$ as $K \rightarrow \infty$.

The quantity $1/D = 1/\sum p_i^2$ is also an *effective number of categories* (under a different weighting than Shannon’s). Sometimes called the “Hill number of order 2.”

How it differs from Shannon. Both measure spread over categories, but they weight rare vs. common categories differently:

- **Shannon** weights rare categories heavily (the $\log p_i$ term diverges as $p_i \rightarrow 0$). A rise in a few new rare categories moves Shannon entropy more.
- **Gini-Simpson** weights common categories heavily (the p_i^2 term). A rebalancing of existing dominant categories moves Gini-Simpson more.

A trend that is picked up by both suggests broad-based diversification. A trend that moves only one suggests something more specific about the shape of the change.

Finite-sample bias. Gini-Simpson’s plug-in bias is smaller than Shannon’s:

$$\mathbb{E}[\widehat{\text{GS}}] - \text{GS}(p) \approx -\frac{1 - \text{GS}(p)}{N}$$

Often acceptable without correction for $N \gtrsim 1000$.

Use in ws2. Secondary per-dimension demographic metric, paired with Shannon entropy. Co-movement of both under trend testing is part of the metric-plurality robustness commitment (desiderata §8).

4 Rao’s Quadratic Entropy

Question answered. Given a categorical distribution *plus* a distance between categories, what is the expected pairwise distance between two random samples from the distribution?

Formal definition. For probabilities $p_1, \dots, p_K$ and a symmetric distance function $d_{ij} \geq 0$ (with $d_{ii} = 0$):

$$Q(p, d) = \sum_{i=1}^K \sum_{j=1}^K p_i p_j d_{ij}$$

Intuition. Rao’s Q generalizes Gini-Simpson. If $d_{ij} = 1$ for all $i \neq j$ (everything is equally distant), $Q = 1 - \sum p_i^2 = \text{Gini-Simpson}$. By specifying non-trivial distances between categories, Q captures the geometric sense in which the distribution is “spread out” in feature space.

Distance choice is load-bearing. The distance metric d_{ij} is a modeling decision. For joint demographic states (e.g., female + Chinese-affiliation + senior + non-top-50-institution), Whitespace 2 uses *weighted Hamming distance*:

$$d_{ij} = \sum_{d=1}^D w_d \cdot \mathbf{1}(\text{state } i \text{ and } j \text{ differ on dimension } d)$$

with uniform weights $w_d = 1/D$ in the primary analysis. Sensitivity analysis doubles one weight at a time.

Why not simpler composites? Plain joint Shannon entropy over joint states treats all category pairs as equally different. For ws2, (female, American, senior) and (male, American, senior) should not be *as* far apart as (female, American, senior) and (male, Chinese, early-career). Rao’s Q captures this; joint Shannon entropy does not.

When it fails.

- Requires a principled distance metric. If the metric is arbitrary, so is the conclusion.
- High-dimensional joint states may have sparse marginals, making $\hat{p}_i$ noisy. Bootstrap CIs are essential.

Use in ws2. Primary composite demographic-diversity metric. Also used for paper-level *team* demographic diversity in Test IV (Section 11).

5 Effective Dimensionality (Participation Ratio)

Question answered. Given a set of N vectors in $\mathbb{R}^d$ (e.g., paper embeddings), how many “independent directions” does the cloud effectively span?

Formal definition. Let the empirical covariance matrix of the N vectors be $\Sigma = (1/N) \sum_{n=1}^N (x_n - \bar{x})(x_n - \bar{x})^\top$, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$. The *participation ratio* (also called effective rank or effective dimensionality) is:

$$d_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} = \frac{(\text{tr } \Sigma)^2}{\text{tr}(\Sigma^2)}$$

Intuition. If all variance is in one direction ($\lambda_1 > 0$, rest zero), then $d_{\text{eff}} = 1$. If variance is spread uniformly across k directions ($\lambda_1 = \dots = \lambda_k = \sigma^2$, rest zero), then $d_{\text{eff}} = (k\sigma^2)^2 / (k\sigma^4) = k$. So d_{eff} counts “how many roughly-equal eigenvalues there are,” which corresponds to how many independent conceptual directions the cloud inhabits.

Why not just count eigenvalues above a threshold? Hard cutoffs are arbitrary and discontinuous. Participation ratio is smooth, scale-invariant (multiplying all eigenvalues by a constant doesn’t change it), and has a clean probabilistic interpretation: it is the exponential of the Rényi-2 entropy of the normalized eigenvalue distribution.

Sample-size dependence. For SPECTER2 embeddings ($d = 768$), estimating d_{eff} requires $N \gtrsim d$ for a stable estimate. The ratio itself converges faster than individual eigenvalues, but small samples bias d_{eff} downward. Whitespace 2 calibrates the minimum sample size empirically in Phase 0.1 sanity Check 5b.

Year-centering. Embedding distributions often have a drifting *mean* (the centroid of CS 1975 sits in a different region than CS 2024), which is a measurement artifact rather than a diversity signal. Whitespace 2 subtracts the year-specific mean before computing d_{eff} , so we measure *spread around center*, not absolute position.

Use in ws2. Primary semantic-diversity metric B (alongside cluster entropy as primary A, mean pairwise cosine distance as secondary).

References. Gao et al. 2017 (neuroscience); Recanatesi et al. 2022 (scale-dependent dimensionality).

6 Cluster Entropy

Question answered. Given embeddings clustered into K groups, what is the Shannon entropy of each year’s distribution over clusters?

Formal definition. Let $c_n \in \{1, \dots, K\}$ be the cluster assignment of paper n . For year Y with papers $\mathcal{P}_Y$:

$$p_k^{(Y)} = \frac{|\{n \in \mathcal{P}_Y : c_n = k\}|}{|\mathcal{P}_Y|}, \quad H_Y = - \sum_{k=1}^K p_k^{(Y)} \log p_k^{(Y)}$$

Apply Miller-Madow bias correction for finite N .

Why cluster over the embedding space. Embedding space is continuous and high-dimensional; a direct Shannon entropy over the continuous distribution requires density estimation, which is expensive and sample-hungry. Clustering discretizes the space into interpretable regions, after which Shannon entropy is straightforward.

The critical methodological commitment: temporal stratification of the cluster fit. The clusters must be fit on a *temporally-stratified* subsample with equal papers per decade, *not* on the full pooled corpus.

If clusters are fit on the unstratified pool, modern papers (which outnumber early-decade papers by 10–50×) dominate the cluster definitions. The clusters then reflect modern semantic structure; old papers map poorly to those modern definitions and concentrate artificially in a few least-wrong clusters, producing spurious low-diversity in early years and a spurious divergence trend. This would silently manufacture the paper’s headline finding as a corpus-composition artifact.

This is desiderata §11 and is load-bearing; it is stress-tested in sanity Check 5b (Phase 0.1) with an A/B comparison quantifying the artifact size on pilot data.

Procedural separation.

- **Fit:** on a temporally-stratified subsample (equal papers per decade).
- **Assign:** every paper in the full corpus (all years) to its nearest cluster centroid from the stratified fit.

Robustness. Report results for $K \in \{30, 50, 100\}$. If the headline trend direction is consistent across K , the finding is robust to cluster-count choice.

Use in ws2. Primary semantic-diversity metric A. Pairs naturally with per-dimension demographic Shannon entropy, enabling direct comparison on the same mathematical quantity (effective-number-of-bins interpretation) in the main 3-panel figure.

7 Citation Gini and Chu-Evans Spearman Top-N

Canonical concentration has two distinct senses:

- (1) **Distributional concentration:** are citations concentrated on a few papers?
- (2) **Temporal persistence:** are the *same* papers at the top year after year?

The actuator-homogenization hypothesis is primarily about (2). Both are captured by Whitespace 2’s primary and secondary metrics.

7.1 Citation Gini (secondary)

The Gini coefficient on citations received in year Y :

$$G_Y = \frac{\sum_{i,j} |c_i - c_j|}{2N^2\bar{c}}$$

where c_i is the number of citations received by paper i in year Y and $\bar{c}$ is the mean. $G = 0$: perfect equality; $G = 1$: one paper gets all citations.

Why “citations received in year Y ” (framing A): this captures the *flow of attention* in year Y , i.e., what the field is drawing on now. The alternative (“citations ever received by papers published in year Y ”) confounds paper age with canonicity. Framing A is a pre-registered commitment and aligns with the Chu-Evans methodology.

7.2 Chu-Evans Spearman top- N correlation (primary)

For each year Y , rank papers by citations received in Y (framing A above). Define the top- N list T_Y . Compute the Spearman rank correlation between T_Y and $T_{Y-\Delta}$:

$$\rho_Y = \text{spearman}(T_Y, T_{Y-\Delta})$$

High $\rho_Y \Rightarrow$ the same papers are at the top year over year $\Rightarrow$ stable canon. Low $\rho_Y \Rightarrow$ canon turns over.

Pre-registered parameters: $N = 50$, $\Delta = 5$. Robustness at $N \in \{100, 500\}$, $\Delta \in \{3, 10\}$.

Why this metric is load-bearing. The subfield mechanism test — “canonical-concentrated subfields show more demographic-semantic divergence” — uses the Chu-Evans Spearman as the canonical- concentration score. This is the paper’s single most important mechanism analysis.

7.3 Why CD-index is deliberately excluded

Park, Leahey & Funk’s CD-index is contested (Petersen-Holst-Macher critiques demonstrated citation-inflation artifacts and dataset extraction bugs). Including CD-index as a primary metric would put ws2 on the wrong side of an active methodological war. Used only for Discussion-section engagement, not measurement.

Part II

Divergence Tests

Whitespace 2 commits to four co-primary divergence tests, each asking a distinct substantive question. They are not redundant; they are complementary.

8 Test I: Standardized-Gap Trend

Question answered. Is the contemporaneous gap between demographic and semantic diversity growing over time?

Construction. Standardize both time series over the analysis window:

$$z_Y^{\text{Dem}} = \frac{\text{DemDiv}_Y - \text{DemDiv}}{\sigma_{\text{DemDiv}}}, \quad z_Y^{\text{Sem}} = \frac{\text{SemDiv}_Y - \text{SemDiv}}{\sigma_{\text{SemDiv}}}$$

Define

$$\text{Gap}_Y = z_Y^{\text{Dem}} - z_Y^{\text{Sem}}$$

Test:

$$\text{Gap}_Y = \alpha + \beta \cdot Y + \epsilon_Y, \quad H_0 : \beta = 0, \quad H_1 : \beta > 0 \text{ (one-sided)}$$

The autocorrelation problem. Time series have serial correlation: ϵ_Y and ϵ_{Y-1} are not independent. Ordinary least squares (OLS) treats them as independent and *underestimates* standard errors, producing false positives.

Newey-West HAC standard errors. The heteroscedasticity-and-autocorrelation-consistent (HAC) variance estimator corrects for this:

$$\widehat{\text{Var}}_{\text{HAC}}(\hat{\beta}) = (X^\top X)^{-1} \hat{\Omega} (X^\top X)^{-1}$$

where

$$\hat{\Omega} = \sum_{t=1}^T \hat{\epsilon}_t^2 x_t x_t^\top + \sum_{l=1}^L w_l \sum_{t=l+1}^T \hat{\epsilon}_t \hat{\epsilon}_{t-l} (x_t x_{t-l}^\top + x_{t-l} x_t^\top)$$

The weights $w_l = 1 - l/(L+1)$ (Bartlett kernel) down-weight far lags. The bandwidth L is typically chosen as $L = \lfloor 4(T/100)^{2/9} \rfloor$ (Newey-West rule) or by bandwidth-selection procedures.

Intuition. Newey-West is OLS with inflated standard errors to account for the fact that correlated residuals give less independent information than a count of data points suggests.

Mann-Kendall as robustness. A nonparametric alternative. Compute the sign statistic:

$$S = \sum_{i < j} \text{sign}(\text{Gap}_{Y_j} - \text{Gap}_{Y_i})$$

Under the null of no trend, S is approximately normal with known variance. No distributional assumption on residuals; robust to outliers. Pre-whitening handles autocorrelation (Hamed & Rao 1998).

Use in ws2. Test I primary on 6 metric pairings per field. Bonferroni-corrected across the 6 pairings within each field.

9 Test II: Confound-Controlled Gap Regression

Question answered. Does the gap between demographic and semantic diversity grow over time *after controlling for known confounds* (field growth, team-size evolution, publishing-norm shifts)?

Construction. Regress Gap on time and controls:

$$\text{Gap}_Y = \beta_0 + \sum_{k=1}^K \beta_k C_{k,Y} + \beta_t Y + \epsilon_Y$$

where C_k are the seven pre-registered controls: $\log N_{\text{papers}}$, trimmed mean team size, median references per paper, arXiv fraction, log distinct active authors, field-entry rate, and subfield composition (vector). Newey-West HAC standard errors.

$$H_0 : \beta_t = 0, \quad H_1 : \beta_t > 0 \text{ (one-sided)}$$

Why not regress SemDiv on DemDiv + time? A prior draft of Test II regressed $\text{SemDiv}_Y = \beta_0 + \beta_1 \text{DemDiv}_Y + \sum \beta_k C_k + \beta_t Y + \epsilon$. That specification has an interpretability problem: if DemDiv itself trends strongly, including Y partials time out of DemDiv’s effect, and β_t can absorb the divergence we want to detect. The Gap-reformulation avoids this by treating divergence as a single quantity and asking whether it trends.

Multicollinearity. The seven controls are all upward-trending over 50 years and are correlated. Variance Inflation Factor (VIF) is reported per variable:

$$\text{VIF}_k = \frac{1}{1 - R_k^2}$$

where R_k^2 is the R^2 from regressing C_k on the other controls. $\text{VIF} > 10$ is a standard flag. Whitespace 2 reports VIF alongside coefficients. Mandatory sensitivity analysis: re-fit Test II (i) with all controls, (ii) with each control dropped in turn, (iii) with subfield composition toggled. β_t stability across these specifications is the robustness claim.

Use in ws2. Test II primary alongside Tests I, III, IV. Directly addresses the “does divergence survive controls?” reviewer question.

10 Test III: De-trended Cross-Correlation and Granger Causality

Question answered. Does demographic diversity predict semantic diversity at some lag k ? If yes, the “divergence” may be *delayed tracking* rather than pure decoupling.

10.1 Cross-correlation at lags

The cross-correlation at lag k between two series X_t, Y_t :

$$\rho_k = \frac{\text{Cov}(X_{t-k}, Y_t)}{\sqrt{\text{Var}(X_t) \cdot \text{Var}(Y_t)}}$$

The spurious-correlation trap. If both X_t and Y_t have strong trends (both rising over time), their raw cross-correlation will be spuriously high — not because they co-vary meaningfully, but because both are going up. A textbook pitfall.

Solution: de-trend first. Two common approaches:

- **First differences:** $\Delta X_t = X_t - X_{t-1}$. Tests whether *changes* in X correlate with *changes* in Y . Simple, aggressive de-trending.
- **Hodrick-Prescott (HP) filter:** extracts a smooth trend component and leaves cyclical residuals. Gentler; less aggressive.

Whitespace 2 uses first differences as primary de-trending and HP-filter residuals as robustness.

Pre-registered lag set. $k \in \{0, 3, 5, 10, 15\}$. Fixed in advance to limit multiple testing. Scanning all lags post-hoc and reporting the largest inflates false-positive rate.

Confidence bands. Under the null hypothesis of independence, ρ_k is approximately normal with variance $1/T$ (Bartlett’s formula for sample sizes T). Confidence bands at $\pm 1.96/\sqrt{T}$. Bootstrap CIs are a robustness option.

10.2 Granger causality

Formalizes “does X predict Y beyond Y ’s own history?” Fit two models:

$$\begin{aligned} \text{Restricted: } Y_t &= \alpha + \sum_{l=1}^L \phi_l Y_{t-l} + u_t \\ \text{Unrestricted: } Y_t &= \alpha + \sum_{l=1}^L \phi_l Y_{t-l} + \sum_{l=1}^L \psi_l X_{t-l} + v_t \end{aligned}$$

F-test: is the improvement from adding X -lags significant?

$$F = \frac{(SSR_R - SSR_U)/L}{SSR_U/(T - 2L - 1)}$$

Caveats.

- Granger causality is *not* causality in the structural sense. It means X has predictive information for Y beyond Y ’s own past. Confounders can manufacture or destroy Granger-causal relationships.
- Lag order L matters. Chosen by AIC (Akaike information criterion) or BIC on pre-pilot data, capped at $L = 5$ in ws2.
- Requires stationarity of both series (after de-trending).

Use in ws2. Test III primary. Four-quadrant interpretation combines Tests I and III:

	Test III: no lagged correlation	Test III: peak at $k > 0$
Test I: significant	true divergence	delayed tracking
Test I: not significant	contemporaneous tracking	lagged tracking

Each quadrant is publishable and tells a different mechanism story.

11 Test IV: Paper-Level Team-Diversity vs. Novelty

Question answered. Do demographically diverse teams produce semantically novel papers?

This is the within-paper companion to the aggregate Tests I–III. It operates at a different level of aggregation (paper cross-section) and addresses the intra-paper version of the claim: *if* diverse inputs get averaged toward canonical outputs by shared actuators, we should see a within-team signature of it.

11.1 Variables

For each paper p :

- **Team diversity** T_p : Rao’s Q over the author team’s joint demographic distribution, with uniform Hamming distance (same metric as the aggregate composite). $T_p = 0$ for single-author papers.
- **Novelty** N_p (**primary**): cosine distance from paper p ’s embedding to the centroid of embeddings of papers p cites:

$$N_p = 1 - \frac{e_p \cdot \bar{e}_{\text{ref}(p)}}{\|e_p\| \cdot \|\bar{e}_{\text{ref}(p)}\|}$$

High $N_p \Rightarrow p$ synthesizes distant concepts; low $N_p \Rightarrow p$ is close to its cited context.

- **Novelty** N_p (**secondary**): cosine distance from p to the year- Y canonical centroid (top-100 most-cited papers in year Y). Tests “distance from canon” as alternative framing.

11.2 Regression

$$N_p = \gamma_0 + \gamma_1 T_p + \sum_k \gamma_k X_{k,p} + \mu_{\text{year}(p)} + \eta_{\text{subfield}(p)} + \epsilon_p$$

where $X_{k,p}$ are paper-level controls (team size, reference count, mean career stage, mean institutional prestige, log paper age, field dummy) and μ, η are year and subfield fixed effects.

Fixed effects, intuitively. A fixed effect absorbs all variation due to year or subfield, leaving only variation *within* year-subfield cells. This answers “within a given year and subfield, do diverse teams produce more or less novel papers?” Year-trends and subfield-differences don’t confound the estimate.

11.3 Standard errors: double-clustered

Papers by the same lead author or in the same subfield are not independent. Double-clustering the standard errors accounts for this:

$$\widehat{\text{Var}}_{\text{DC}}(\hat{\gamma}_1) = V_{\text{author}} + V_{\text{subfield}} - V_{\text{author} \cap \text{subfield}}$$

(Cameron, Gelbach & Miller 2011 estimator). Naïve OLS standard errors that ignore clustering underestimate variance and produce false positives.

11.4 Hypotheses and effect size

- H_0 : $\gamma_1 = 0$ (team diversity uncorrelated with novelty)
- H_1 (actuator-dominant): $\gamma_1 < 0$ — diverse teams produce *less* novel work; shared actuators average diverse inputs toward canonical outputs.
- H'_1 (diversity-productive): $\gamma_1 > 0$ — diverse teams produce more novel work. Consistent with Hofstra et al. 2020 framing.

Two-sided test; the *sign* of γ_1 is the substantive answer. With hundreds of thousands of papers, statistical significance is almost automatic, so substantive-significance threshold $|\gamma_1| \geq 0.05$ (after standardization of T_p and N_p) is pre-registered.

Time interaction (secondary). $\gamma_1(Y) = \gamma_{10} + \gamma_{11}Y$: does the relationship evolve? A strengthening negative γ_1 over time would be consistent with growing actuator dominance.

Causal caveat. Cross-sectional correlations $\neq$ causal effects. γ_1 is reported as descriptive association. Reverse causation (novel work attracts diverse teams), selection (risk-prefering researchers form diverse teams), and omitted variables all threaten causal interpretation. No instrumental variable strategy is attempted in Whitespace 2.

Part III

Break-Point Analysis

If Tests I or II detect significant divergence, a natural follow-up is: when does it start? A continuous slow divergence from 1970 tells a different mechanism story than a sharp break around, say, 2000.

12 Chow Test at Pre-specified Candidates

Setup. Suppose a candidate break year τ is specified in advance (e.g., 2000, corresponding to the dot-com peak). Fit two regressions of Gap on year:

$$\begin{aligned}\text{Pre: } \text{Gap}_Y &= \alpha_1 + \beta_1 Y + u_Y, & Y \leq \tau \\ \text{Post: } \text{Gap}_Y &= \alpha_2 + \beta_2 Y + v_Y, & Y > \tau \\ \text{Pooled: } \text{Gap}_Y &= \alpha + \beta Y + w_Y, & \text{all } Y\end{aligned}$$

Let SSR_P, SSR_1, SSR_2 be the residual sums of squares. The Chow F-statistic:

$$F = \frac{(SSR_P - SSR_1 - SSR_2)/2}{(SSR_1 + SSR_2)/(T - 4)}$$

tests the null that coefficients are equal on both sides of τ .

Pre-specification is the key. Post-hoc search for break points inflates false positives enormously — with 50 years of data, *some* year will look like a break by chance. Chow requires the candidate to be committed in advance. Whitespace 2 pre-registers specific candidates per field, tied to substantive events.

Pre-registered candidates (CS):

- 1991–93: arXiv launch + public web
- 1998–2000: dot-com peak (demographic broadening from industry demand)
- 2008–09: financial crisis, academic-market consolidation
- 2012: AlexNet, deep-learning takeoff
- 2018–20: foundation models (likely too recent for smoothed series)

Pre-registered candidates (Physics):

- 1991: arXiv launched in hep-th
- 1995–2000: post-SSC cancellation; string theory dominance
- 2012: Higgs boson discovery

Bonferroni correction across the candidate set within each field.

13 Bai-Perron Multi-Break Detection

Setup. Rather than pre-specifying candidates, the Bai-Perron procedure algorithmically searches for m breakpoints $\tau_1 < \tau_2 < \dots < \tau_m$ partitioning the time series into $m + 1$ regimes, each with its own slope and intercept. The number of breaks m is selected by BIC or sequential F-tests.

Trimming. Breaks cannot be detected near series boundaries (too few points on one side for stable estimates). Standard trimming removes the first and last 15% of the series. For 1970–2024, this excludes break detection before ~ 1978 or after ~ 2017 .

Confidence intervals on break locations. Bai-Perron provides asymptotic CIs on each τ_i . Bootstrap CIs are a robustness option.

Power. With annual data over 55 years, the Bai-Perron procedure can reliably detect 0 or 1 breaks; 2 breaks is borderline; 3+ is generally not defensible at this series length. Whitespace 2 reports what BIC selects plus sensitivity at ± 1 break.

Concordance requirement. A break is “significant” only if *both* Bai-Perron detects it *and* Chow at a nearby pre-specified candidate (within ± 2 years) rejects no-break. Prevents one-method-only artifacts from being reported as substantive.

Part IV

Auxiliary Methods

14 Bootstrap Confidence Intervals

What and why. Classical confidence intervals rely on distributional assumptions (usually normality) about the sampling distribution of the statistic. For complex statistics like Rao’s Q or participation-ratio effective dimensionality, these assumptions may not hold. The bootstrap sidesteps the issue.

Algorithm. Given data $\mathcal{D} = \{x_1, \dots, x_N\}$ and statistic $\hat{\theta}(\mathcal{D})$:

1. For $b = 1, \dots, B$: draw a sample $\mathcal{D}^{*(b)}$ of size N with replacement from $\mathcal{D}$; compute $\hat{\theta}^{*(b)} = \hat{\theta}(\mathcal{D}^{*(b)})$.
2. Form the empirical distribution of $\{\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(B)}\}$.
3. For 95% CI: take the 2.5th and 97.5th percentiles (percentile method) or the bias-corrected equivalent (BCa method).

Commitment. Whitespace 2 uses $B = 200$ replicates for every metric on every figure. 95% percentile CIs unless otherwise noted.

When it fails.

- Small sample sizes ($N < 30$) give noisy bootstrap estimates.
- Statistics at distributional extremes (maxima, minima) are poorly bootstrapped. Not an issue for our metrics.
- Time-series bootstrap requires block bootstrapping (Politis-Romano) to preserve autocorrelation structure. For our annual series in the divergence tests, we use moving-block bootstrap when computing CIs on trend estimates.

15 Multiple-Comparisons Correction

The problem. Running many statistical tests inflates the probability of at least one false positive. With 100 tests at $\alpha = 0.05$ under the global null, we expect 5 false positives just by chance.

15.1 Bonferroni correction

Divide α by the number of tests:

$$\alpha_{\text{Bonf}} = \frac{\alpha}{m}$$

Controls the *family-wise error rate* (FWER): probability of *any* false positive among the m tests. Conservative; loses power when tests are correlated (since true effective m is smaller).

15.2 False discovery rate (FDR)

Benjamini-Hochberg procedure. Sort p-values $p_{(1)} \leq \dots \leq p_{(m)}$. Reject tests $p_{(i)}$ where $p_{(i)} \leq (i/m) \cdot q$ for target FDR level q (often 0.05 or 0.10).

Controls the *expected proportion* of false discoveries among rejections. Less conservative than Bonferroni; more powerful when many true effects exist.

15.3 Whitespace 2’s hierarchical scheme

- Primary tests (Tests I–IV, 6 metric pairings each, 2 fields): Bonferroni within each test-type across 6 pairings; *no correction across test-types* since they ask distinct questions. Replicated divergence requires all 3 test types to agree on direction in $\geq 4/6$ pairings per field.
- Secondary tests (per-dimension demographic decomposition: 7 dims $\times$ 3 semantic $\times$ 2 fields = 42 tests): FDR at $q = 0.10$, labeled exploratory.
- Break-point analysis: Bonferroni within candidate set per field.

16 Procrustes Alignment

Problem setup. In the Flavor A drift mitigation, we train separate embedding spaces per era and then need to *align* them so embeddings from different eras can be compared. Each space is defined up to an arbitrary rotation/reflection.

Orthogonal Procrustes problem. Given two matrices $W_1, W_2 \in \mathbb{R}^{k \times d}$ (each row a vector), find the orthogonal matrix R minimizing:

$$R^* = \arg \min_{R: R^\top R = I} \|W_1 R - W_2\|_F$$

Closed-form solution via SVD. Compute $W_2^\top W_1 = U \Sigma V^\top$. Then $R^* = V U^\top$.

The anchor-paper idea. To align era 1 (say, 1980s embeddings) with era 2 (2010s embeddings), we need vectors that mean the “same thing” in both eras. These are anchor papers or anchor concepts appearing in both eras. Their embeddings in era 1 form W_1 ; their embeddings in era 2 form W_2 . Procrustes finds the rotation aligning these anchors. Then apply the same rotation to all era-1 embeddings to bring them into era-2’s coordinate system.

Anchor-choice caveat. Whatever makes it into the anchor set defines the “fixed frame.” Anchors should be semantically stable across eras (named methods, classical theorems, widely-agreed terms). This is itself a modeling commitment; Whitespace 2 treats anchor selection as a Phase 0.2 deliverable.

17 Variance Inflation Factor (VIF)

Problem. In regression with multiple controls, if controls are highly correlated, coefficient estimates become unstable and their standard errors explode. This is multicollinearity.

Diagnostic. For control k , regress it on the other controls and record R_k^2 . Define:

$$\text{VIF}_k = \frac{1}{1 - R_k^2}$$

$\text{VIF}_k = 1$: control k is uncorrelated with others. $\text{VIF}_k = 10$: $R_k^2 = 0.9$; 90% of k ’s variance is explained by other controls. Standard error of $\hat{\beta}_k$ is inflated by a factor $\sqrt{\text{VIF}_k}$.

Rules of thumb.

- VIF < 5: unproblematic.
- VIF 5–10: concerning; re-examine.
- VIF > 10: drop, combine, or report stability under omission.

Whitespace 2 treatment. The seven Test II controls are all upward-trending over 50 years; high VIFs are *expected*. Rather than dropping variables mechanically, ws2 commits to a *sensitivity analysis*: re-fit with each control dropped in turn, report β_t across specifications. Qualitative stability of β_t is the robustness claim.

18 Missing Data: MAR, MNAR, and Inverse-Probability Weighting

Problem. Whitespace 2’s analytical population P — papers with non-empty `abstract_inverted_index` in OpenAlex — is $\sim 50\%$ of the OpenAlex CS+Physics field. Aggregate metrics computed on P are biased estimates of the field-level aggregates if the selection mechanism (whether a paper has an abstract or not) correlates with the metric being measured. Phase 0.1 Check 1f confirmed this is the case along three axes: citation count, concept distribution, and country of first authorship — all aligned with ws2’s three substantive measurement axes.

Rubin’s missing-data taxonomy (Rubin 1976; standard in econometrics, biostatistics, and survey statistics). Let D be the indicator that a paper has an abstract (1 if observed, 0 if missing), let X be the observed covariates (year, field, type, citation tertile, concept cluster, known country), and let Y be the outcome of interest (e.g., paper-level semantic novelty, demographic attribute, citation rank).

- **MCAR (Missing Completely At Random):** $\Pr(D = 1 \mid X, Y) = \Pr(D = 1)$. Missingness is independent of everything observable and unobservable. Aggregate means are unbiased without correction. *ws2 does not assume MCAR*; Check 1f rules it out empirically.
- **MAR (Missing At Random):** $\Pr(D = 1 \mid X, Y) = \Pr(D = 1 \mid X)$. Missingness depends on *observed* covariates but not on the missing value itself, conditional on those covariates. Two papers with identical X have the same probability of being observed regardless of what their outcome Y would have been. **Inverse-probability-of-availability weighting recovers full-population aggregates exactly under MAR.**
- **MNAR (Missing Not At Random):** $\Pr(D = 1 \mid X, Y) \neq \Pr(D = 1 \mid X)$. Missingness depends on the unobserved Y itself, even after conditioning on X . Recovered aggregates are biased; we can only *bound* the population aggregate, not point-identify it.

Inverse-probability weighting under MAR. For each paper $p \in P$, fit a propensity model $\hat{\pi}(X_p) = \Pr(D = 1 \mid X = X_p)$ and define the weight $w_p = 1/\hat{\pi}(X_p)$. The weighted aggregate

$$\hat{M}_{\text{corrected}} = \frac{\sum_{p \in P} w_p m_p}{\sum_{p \in P} w_p}$$

is an unbiased estimate of the OpenAlex full-population aggregate M under MAR. Intuition: papers with low $\hat{\pi}(X_p)$ are underrepresented in P (papers like them are rarely observed); their weights are correspondingly large to compensate. Under MAR, the conditional distribution of Y given X is the same in P as in the full population, so re-weighting by $1/\hat{\pi}$ recovers the population distribution.

Sensitivity to MNAR. The MAR assumption is testable in principle but never directly verifiable (we cannot observe Y for $D = 0$ papers). The standard approach is a *sensitivity grid*: parametrize a

tilt $\gamma \in [-\Gamma, \Gamma]$ representing how much the missingness mechanism deviates from MAR, recompute the bound $\hat{M}(\gamma)$ for a range of γ , and report the bound $[\hat{M}(-\Gamma), \hat{M}(+\Gamma)]$. ws2 commits to a pre-registered Γ -grid in Phase 0.2; tentative Γ corresponds to $\pm 50\%$ selection-probability variation within X -strata.

Reliability bound on the propensity model. The inverse-probability-weighted estimate is only as good as the propensity model. Pre-registered diagnostics:

- **Held-out CV-AUC** of the fitted propensity model. ws2 commits to a pre-registered threshold (tentative ≥ 0.75 in Phase 0.2). Below threshold, the corrected aggregate is not the headline; uncorrected + MNAR-tilt bound is reported.
- **Calibration plot** of predicted vs. observed $\hat{\pi}$ across deciles. Severe miscalibration indicates the propensity model is misspecified; correction may be unreliable even at adequate AUC.
- **Stabilized weights** $w_p/\bar{w}$ to flag high-leverage observations (papers with extreme weights drive the aggregate; their over-influence inflates variance).
- **Functional-form sensitivity:** report aggregates under logistic regression (baseline) and gradient-boosted trees (flexibility check). Substantial disagreement is itself a diagnostic of nonlinear interactions the baseline misses.

Whitespace 2 treatment. §9e in the Phase 0.1 plan operationalizes inverse-probability weighting on $X = \text{year} \times \text{field} \times \text{type} \times \text{citation_tertile} \times \text{concept_cluster} \times \text{known_country}$. Headline aggregates are reported with both the MAR-corrected estimate (when CV-AUC clears the threshold) and the MNAR-tilt bound (always). The country axis is not propensity- correctable (the UNKNOWN bucket has no observable strata to fit on); demographic-plurality claims are scope-narrowed to P_{demo} ($\sim 50\%$ with determinable country) by construction. See plan §9e for the full operational specification and §0 for the analytical-population definitions.

Why MAR/MNAR is the right framework here. Selection bias caused by missing data is structurally distinct from attribute-classification error (e.g., NamSor misgendering authors) and from metric-construct invalidity (e.g., CD-index zero-reference artifacts). The three are orthogonal construct-validity threats; ws2’s Methods overview (§14) frames them as four parallel audits. The MAR/MNAR taxonomy is the principled toolkit for the analytical-population audit; without it, “selection bias” is too fuzzy to bound and too tempting to wave away.

Appendix

Glossary of Symbols

Symbol	Meaning
$H(p)$	Shannon entropy of distribution p
$GS(p)$	Gini-Simpson index
$Q(p, d)$	Rao’s quadratic entropy with distance d
d_{eff}	Effective dimensionality (participation ratio)
λ_i	i -th eigenvalue of covariance matrix
$p_k^{(Y)}$	Cluster- k probability in year Y
G_Y	Citation Gini in year Y
ρ_Y	Chu-Evans Spearman top- N at year Y , lag Δ
$z_Y^{\text{Dem}}, z_Y^{\text{Sem}}$	Standardized demographic, semantic diversity
Gap_Y	$z_Y^{\text{Dem}} - z_Y^{\text{Sem}}$
β_t	Slope on year in Test II
T_p	Team demographic diversity of paper p
N_p	Novelty of paper p
γ_1	Coefficient of T_p in Test IV regression
μ, η	Year and subfield fixed effects
τ	Break-point year
VIF_k	Variance inflation factor for control k
B	Bootstrap replicates

Key References by Method

- **Shannon entropy, Miller-Madow correction:** Miller 1955 “Note on the bias of information estimates.”
- **Rao’s Q:** Rao 1982, “Diversity and dissimilarity coefficients: a unified approach,” *Theor. Popul. Biol.*
- **Participation ratio:** Gao et al. 2017 (neural dimensionality); Recanatesi et al. 2022.
- **Chu-Evans Spearman:** Chu & Evans 2021, *PNAS* 118(41).
- **Newey-West HAC:** Newey & West 1987, *Econometrica*.
- **Mann-Kendall with pre-whitening:** Hamed & Rao 1998, *J. Hydrology*.
- **Granger causality:** Granger 1969, *Econometrica*.
- **Bai-Perron:** Bai & Perron 1998, *Econometrica* 66(1); 2003, *J. Appl. Econ.*
- **Cameron-Gelbach-Miller double-clustering:** Cameron, Gelbach & Miller 2011, *J. Bus. Econ. Stat.*
- **Bootstrap:** Efron & Tibshirani 1993, *An Introduction to the Bootstrap*.
- **Block bootstrap for time series:** Politis & Romano 1994, *JASA*.
- **FDR:** Benjamini & Hochberg 1995, *J. R. Stat. Soc. B*.
- **Orthogonal Procrustes:** Schönemann 1966, *Psychometrika*.
- **Diachronic word embeddings:** Hamilton, Leskovec & Jurafsky 2016, *ACL*.

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```
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```

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